

Vaibhav Kanojia

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Education

Delhi Technological University (DCE)
Bachelor of Technology in Computer Science

August 2023 – May 2027
CGPA: 8.62

ASN Senior Secondary School, Delhi
Class 12, CBSE

2023
Percentage: 95.6%

ASN Senior Secondary School, Delhi
Class 10, CBSE

2021
Percentage: 95.5%

Work Experience

Software Engineering Intern | OpenTrade | Y Combinator S26
Reporting to CEO & CTO

July 2026 – Present

- First engineering hire at a **Y Combinator (S26)** company; conceptualized and shipped the platform's first **in-app game** from zero as the sole engineer, reporting directly to both co-founders.
- Built a multi-source **AI content pipeline** integrating news, viral, and social feeds across **5+ platforms**; engineered a **hierarchy-based dispatcher** with per-event impact scoring – threshold-gated **real-time delivery** vs. batched updates – eliminating redundant live calls while keeping critical events sub-second.
- Instrumenting click-streams, reaction patterns, and session signals across **1,000+ daily active users** to map virality drop-off; iterating on **in-app games and gamification mechanics** to push **K-factor** above **1.0** and drive organic growth.

Publications & Research

Divide and Translate: Parameter Isolation with Encoder Freezing for Low-Resource Indic NMT | [Link](#)

IJCNLP-AACL 2025

- Proposed a parameter-isolation framework using a **frozen NLLB-600M encoder coupled with 8 direction-specific decoders** to eliminate negative transfer in ultra-low-resource Indic translation while preserving cross-lingual semantic representations.
- Achieved leaderboard scores of **171.4 / 161.1** with **less than 6% public-private drift** across Hindi-Bhili, Hindi-Gondi, Hindi-Mundari and English-Santali, evaluated under a weighted BLEU-chrF protocol on the MMLoSo 2025 benchmark.

Projects

GrantWriter – NIH SBIR Phase I Grant Writing Platform | [Live](#) | [GitHub](#) | RAG Pipeline, LLM Orchestration

- Built for a **Y Combinator-backed startup**: two-pass **LLM pipeline** (coarse-draft to refine to cite) grounded by **pgvector** cosine search over **2,705 won NIH R43 grants**; **deterministic compliance linter** gates PDF export against the real NIH solicitation requirements.
- Full-stack production deployment: **Next.js**, **Drizzle ORM**, **Anthropic API**, **Voyage AI** embeddings, **Stripe** payments, **Clerk** auth; **5-case eval harness** with rubric scorer and **CI gate at 80% pass rate** before any output reaches the user.

Project Synapse – Autonomous agent for last-mile delivery disruptions | [Link](#) | Agentic AI, Multi-Step Reasoning

- Designed a custom **ReAct loop** (perceive, plan, execute, reflect, close) over a **typed tool registry**, exposing function signatures to the LLM and orchestrating multi-tool resolution plans for last-mile delivery disruptions.
- Built **structured trace logging** for every plan, tool call, and reflection step, with a **3-tier provider fallback** that degrades gracefully to a deterministic mock – the agent runs end-to-end with no internet, no API keys, and no LangChain.

GitGalaxy – Explore GitHub repo as navigable 3D semantic galaxy | [Link](#) | 3D Knowledge Graph, Semantic Search

- Used **jina-embeddings-v2-base-code** to generate 768-dim representations for every typed symbol extracted by **Tree-sitter**, then projected the resulting manifold into 3D with **UMAP** for spatial exploration in WebGL.
- Tuned a **FAISS index** delivering **sub-20ms queries** over the embedded corpus, backed by a content-addressed cache that loads pre-computed repos instantly; runs fully on-device with no API keys.

Skills

Programming Languages: Python, C, C++

ML / Systems Stack: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

Certifications: Machine Learning & Deep Learning (Stanford), Intro & Intermediate ML (Kaggle)

Achievements & Extracurriculars

4× National Hackathon wins: 1st @ Quantum Hackathon – IIT Jodhpur | 1st @ Hack The Universe – Delhi University | 2nd @ TensorQuest – IEST Shibpur | 3rd @ IMPULSE 2025 – NIT Karnataka.

Top 1% in Amazon ML Challenge 2025 (rank **67** / **20,000+**) and Kaggle Playground Series Oct 2024 (rank **22** / **3,858**).

Top 2% in the Wunder Fund Quant Challenge – rank **79** / **5,000**, within **0.015** of the global leader.

Director @ **Neural AI Society** | Researcher with **Calibre DTU** | IEEE Student Member